

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO1: *Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems.(BL2,BL3)*

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10 Total Marks: 50 Annexure A

Case Study Duration: 2hrs

1. A logistics company is developing an RL-based warehouse robot that learns to pick, pack, and transport items efficiently while avoiding obstacles and minimizing delays. Analyze how the components of a Markov Decision Process (states, actions, rewards, policy, and environment) can be defined for this system. Design a reward function that balances speed, accuracy, and safety, and evaluate how reward design influences the robot's learning and behavior. **(5 Marks)**
2. A streaming platform wants to use Reinforcement Learning to recommend movies dynamically based on user preferences and viewing history. Apply RL concepts to construct a suitable state representation, action space, and reward function. Analyze how different reward signals affect user engagement and evaluate the trade-off between exploration and exploitation. **(5 Marks)**
3. An autonomous agricultural system uses RL to optimize irrigation and fertilizer usage based on soil conditions and weather forecasts. Design an RL framework by defining states, actions, and rewards. Analyze the effect of delayed rewards on crop yield optimization and justify the choice of a suitable learning algorithm. **(5 Marks)**
4. A cybersecurity system uses RL to detect and respond to network attacks in real time. Analyze how states and environment can be modeled for intrusion detection. Design appropriate actions and reward functions, and evaluate the challenges of sparse rewards and real-time decision-making. **(5 Marks)**
5. A ride-sharing company plans to use RL to optimize driver allocation and dynamic pricing strategies. Apply RL concepts to define states, actions, and rewards. Analyze how environmental uncertainty affects learning and evaluate ethical concerns related to pricing strategies. **(5 Marks)**
6. A smart home automation system uses RL to control lighting, heating, and cooling based on occupancy and weather conditions. Design an RL framework including states, actions, and reward function. Analyze how conflicting objectives like comfort and energy saving can be handled and evaluate the impact of reward shaping. **(5 Marks)**
7. A gaming company is developing an RL-based agent to play a complex strategy game against human players. Analyze how the problem can be modeled as an MDP. Design a reward structure that encourages long-term strategy and evaluate how exploration strategies affect performance.

(5 Marks)

8. A financial institution uses RL to detect fraudulent transactions in real time. Apply RL principles to define the state space, action space, and reward function. Analyze the challenges of imbalanced data and delayed feedback, and evaluate the risks associated with incorrect decisions.

(5 Marks)

9. A healthcare system uses RL to recommend personalized rehabilitation exercises for patients based on their progress. Design an RL framework with suitable states, actions, and rewards. Analyze the impact of delayed rewards and patient variability, and evaluate ethical concerns and safety considerations.

(5 Marks)

10. A smart grid system uses RL to manage electricity distribution and reduce peak load demand. Analyze how the components of an MDP can be defined in this context. Design a reward function to balance efficiency and reliability, and evaluate the challenges of scalability and real-time learning.

(5 Marks)

Code of Conduct:

I A Afran / 192325024 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CASE STUDY

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided

		justification			
Clarity	10	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Question 6 – Answer

Problem Statement

A smart home automation system uses RL to control lighting, heating, and cooling based on occupancy and weather conditions. Design an RL framework including states, actions, and reward function. Analyze how conflicting objectives like comfort and energy saving can be handled and evaluate the impact of reward shaping. (5 Marks)

Title

Design of a Reinforcement Learning Framework for Smart Home Automation using Occupancy and Weather

Conditions. **1. Introduction**

1.1 Introduction

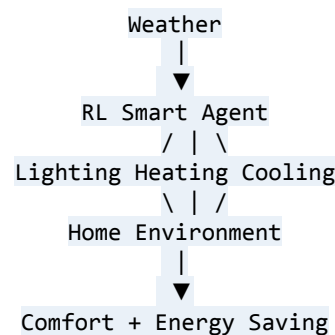
Smart home automation is the process of automatically controlling household devices such as lighting, heating, ventilation, and air conditioning (HVAC). Traditional automation systems rely on fixed schedules or manually defined rules, which often fail to adapt to changing environmental conditions or user behavior.

Reinforcement Learning (RL) enables the smart home to learn optimal control strategies by interacting with the environment. An RL agent continuously observes occupancy, weather conditions, and indoor temperature, then decides the best actions to improve comfort while reducing energy consumption.

By receiving rewards based on its decisions, the agent gradually learns an energy-efficient policy that balances user satisfaction and operational cost.

1.2 Objectives

- Design an RL framework for smart home automation.
- Identify states, actions, and rewards.
- Improve occupant comfort.
- Reduce electricity consumption.
- Analyze conflicting objectives.
- Study the impact of reward shaping.



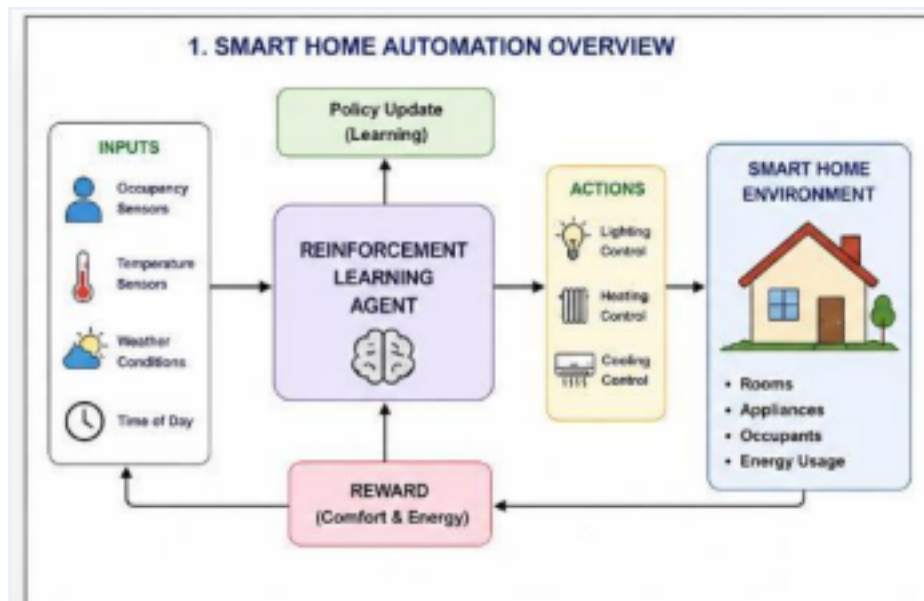


Figure 1: Smart Home Automation Overview

2. Reinforcement Learning Framework

Agent

The intelligent controller that decides how to operate the appliances. Examples:

- Turn lights ON/OFF
- Increase room temperature
- Reduce cooling

Environment

The smart home itself, including:

- Rooms
- Occupants
- Temperature
- Weather
- Electrical appliances

States

The state represents the current condition of the house. Example state

- variables:
- Occupancy
 - Indoor temperature
 - Outdoor temperature
 - Weather
 - Time of day
 - Energy usage

Actions

The agent can perform actions such as:

- Turn light ON
- Turn light OFF

- Increase heater
- Decrease heater
- Increase cooling
- Decrease cooling
- Maintain current settings

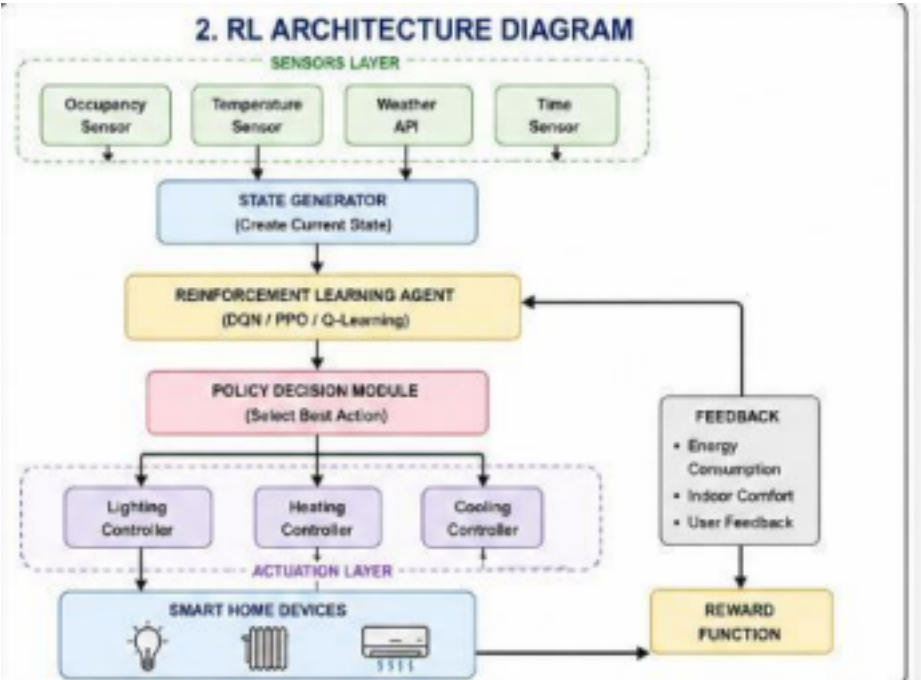


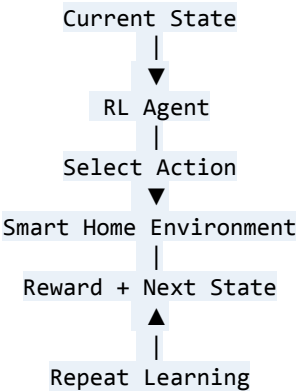
Figure 2: RL Architecture Diagram

3. Rewards

The environment provides rewards based on the chosen action.

Reward Type	Example Conditions	Value
Positive reward	Comfort maintained; Energy reduced	+10
Negative reward	Room too hot; Lights ON in empty room	-8

RL Interaction Diagram



Workflow

1. Sensors collect data.
2. State is generated.
3. RL agent observes state.
4. Action is selected.
5. Devices operate.
6. Energy and comfort are measured.
7. Reward is calculated.
8. Agent updates policy.

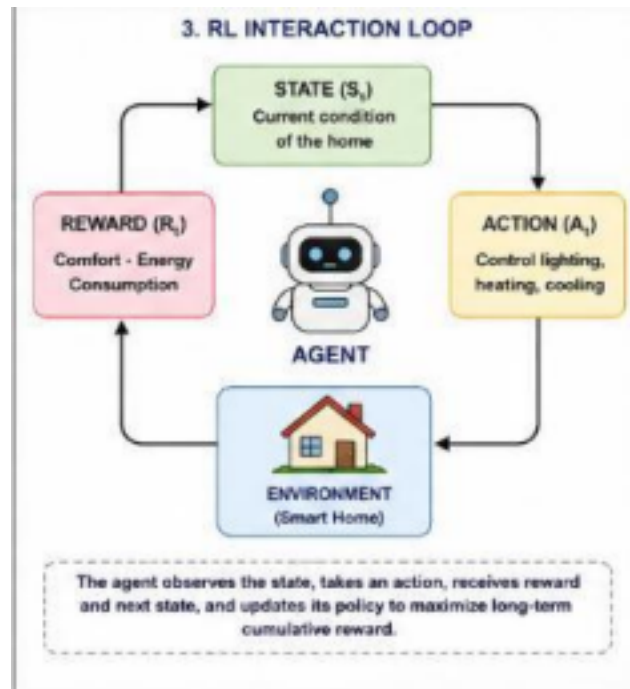


Figure 3: Interaction Loop

4. Reward Function

A simple reward function can be written as:

$$\text{Reward} = \text{Comfort Score} - \text{Energy Consumption}$$

Where:

- Comfort Score increases when indoor conditions match user preferences.
 - Energy Consumption decreases the reward when excessive electricity is used.
- This encourages the agent to maximize comfort while minimizing power usage.

5. Conflicting Objectives

The biggest challenge is balancing comfort and energy saving.

Scenario 1

- Winter.
- Occupant is inside.
- Desired temperature = 24°C.
- The RL agent increases heating.
- Comfort is high.

- Energy consumption also increases.

Scenario 2

- Nobody is home.
- The RL agent turns OFF lights.
- Heating is reduced.
- Energy saving increases.
- Comfort is not affected because no one is present.

Conflict Table

Situation	Comfort	Energy Saving	RL Decision
Occupied room	High	Medium	Maintain comfort
Empty room	Low priority	High	Turn devices OFF
Hot weather	Medium	Medium	Moderate cooling
Cold weather	High	Medium	Moderate heating

Handling Conflicts

The RL agent learns priorities through the reward function:

- Positive rewards for maintaining comfortable conditions.
- Negative rewards for unnecessary energy usage.
- Balanced decisions based on occupancy and weather.



Figure 4: States, Actions and Reward

6. Impact of Reward Shaping

Reward shaping provides additional guidance to help the RL agent learn faster.

Without Reward Shaping

- Learning is slower.
- The agent explores many poor actions.
- Higher energy wastage during training.

With Reward Shaping

- Faster convergence.
- Improved comfort.
- Reduced energy consumption.
- More stable decision making.

Comparison Table

Parameter	Without Reward Shaping	With Reward Shaping
Learning Speed	Slow	Fast
Energy Saving	Medium	High
Occupant Comfort	Moderate	High
Policy Stability	Low	High

Advantages

- Lower electricity bills.
- Better indoor comfort.
- Intelligent adaptation to weather.
- Reduced human intervention.
- Sustainable energy usage.

Conclusion

This case study demonstrated the design of a Reinforcement Learning framework for smart home automation. The RL agent learns from occupancy, weather, and temperature data to control lighting and HVAC systems efficiently. By carefully designing states, actions, and reward functions, the system balances user comfort with energy efficiency. Reward shaping further improves learning speed and overall performance.

Future Scope

- Integration with solar energy systems.
- Voice-controlled automation.
- Multi-room optimization.
- Deep Reinforcement Learning (DQN, PPO).
- Smart grid integration.
- IoT-enabled predictive control.